

Econ 381: Introduction to Econometrics

Handout 6: Inference in Simple Linear Regression

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Where Handout 5 left us

For the California school districts, we fitted

$$\widehat{\text{TestScore}} = 698.93 - 2.28 \text{ STR}.$$

Handout 5 explained why a different random sample gives a different slope.

The superintendent's next question

How much uncertainty surrounds -2.28 ? Could sampling variation alone explain a slope this far from zero?

We will estimate that uncertainty from the same sample, then use it to construct a range of plausible slopes and evaluate a hypothesis.

As before, a causal interpretation requires assumptions about omitted determinants of achievement (regression error). A regression table alone cannot determine them.

Roadmap

1. **Section 5.1: standard error of $\hat{\beta}_1$.** Estimate how much slopes vary across samples.
2. **Section 5.2: confidence intervals.** Describe uncertainty about β_1 .
3. **Section 5.1: hypothesis testing.** Read the CI, t -statistic, and p -value together.
4. **Section 5.3: binary regressors.** Interpret a slope as a difference in means.
5. **Section 5.4: heteroskedasticity and homoskedasticity.** Understand why we use robust standard errors.

We continue the California schools example and the 3,000-student experiment.

Section 5.1

The standard error of $\hat{\beta}_1$

One student's sample must measure its own uncertainty

Imagine 3,000 students each estimating a slope from a random sample.

- ▶ The **standard deviation of the slopes** describes their spread across all possible samples of a fixed size.
- ▶ One student sees just one sample and one slope.
- ▶ A **standard error** uses that one sample to estimate the spread the student would see across repeated samples.

The distinction

$SD(\hat{\beta}_1)$ is a fixed population quantity for a given n .

$SE(\hat{\beta}_1)$ is an estimate that changes from sample to sample.

Recall: *SER* measures the scatter of *outcomes* around the fitted line. It does not measure the sampling uncertainty of the slope.

The sample mean and the slope have the same logic

Under the assumptions from Handouts 4 and 5, in large samples:

$$\bar{Y} \stackrel{\text{approx.}}{\sim} N\left(\mu_Y, \frac{\text{Var}(Y)}{n}\right),$$

$$\hat{\beta}_1 \stackrel{\text{approx.}}{\sim} N\left(\beta_1, \frac{\text{Var}(v_i)}{n[\text{Var}(X)]^2}\right), \quad v_i = (X_i - E[X])u_i.$$

- ▶ Each estimator is approximately centered at its fixed target.
- ▶ Each variance is a population expression divided by sample size n .
- ▶ Each standard deviation has a factor $1/\sqrt{n}$.

For the slope: A1 $E[u_i | X_i] = 0$; A2 i.i.d. observations; A3 large outliers are rare (finite fourth moments), plus $\text{Var}(X) > 0$ (from the bonus part in the midterm exam).

The unknown pieces in the slope's standard deviation

$$\text{SD}(\hat{\beta}_1) \approx \frac{1}{\sqrt{n}} \frac{\sqrt{\text{Var}[(X_i - E[X])u_i]}}{\text{Var}(X)}.$$

The formula contains quantities we do not know:

1. The population line, hence each error u_i .
2. The population mean $E[X]$ and both population variances.

Our construction

1. Replace population regression errors with sample residuals.
2. Replace population expectations and variances with sample counterparts.
3. Take the square root of the variance estimate and divide by $\sqrt{n}$.

This follows Handout 4, where $\text{SD}(\bar{Y}) = \sqrt{\text{Var}(Y)}/\sqrt{n}$ led to $\text{SE}(\bar{Y}) = s_Y/\sqrt{n}$.

Step 1: replace regression errors with residuals

$$\underbrace{u_i = Y_i - \beta_0 - \beta_1 X_i}_{\text{distance from the population line}} \quad , \quad \underbrace{\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i}_{\text{distance from the sample line}} \quad .$$

Subtract the two:

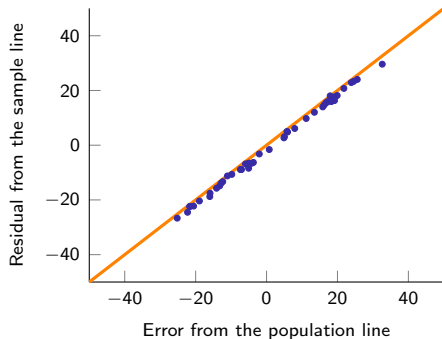
$$\hat{u}_i - u_i = -(\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1)X_i.$$

When the estimated coefficients are close to the population coefficients, the two vertical distances are close at a given X_i .

Why this helps

We can calculate every residual. If estimated slopes and intercepts are close to their population values, then we can replace unknown regression errors with known residuals.

Errors and residuals in the school-data experiment



Treat all 420 districts as our teaching population. We know its best-fitting line and each district's error from that line.

Each student randomly draws 50 districts with replacement and fits a sample line. That student can compute residuals from the sample line.

Points near the orange 45° line have similar errors and residuals. They need not be identical.

Source: Stock and Watson, `caschool1.dta`, 420 California school districts.

Step 2a: estimate the variance of X

The denominator contains

$$\text{Var}(X) = E[(X - E[X])^2]$$

Estimate it with the familiar sample variance:

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

- ▶ Replace the unknown $E[X]$ by the observed $\bar{X}$.
- ▶ Calculate each squared distance from that sample average.
- ▶ Add these distances and divide by $n - 1$.

The variance of the slope contains $[\text{Var}(X)]^2$. After taking the square root, the denominator of its *standard deviation* is $\text{Var}(X)$.

Step 2b: estimate the variance of v_i

Handout 5 defined $v_i = (X_i - E[X])u_i$. Replacing the unobserved u_i and $E(X)$ with their sample counterparts $\hat{u}_i$ and $\bar{X}$, respectively, gives

$$\hat{v}_i = (X_i - \bar{X})\hat{u}_i.$$

Similar to the residual average, OLS with an intercept implies

$$\bar{\hat{v}} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})\hat{u}_i = 0,$$

and the sample variance simplifies to

$$s_{\hat{v}}^2 = \frac{1}{n-1} \sum_{i=1}^n (\hat{v}_i - 0)^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2.$$

An observation contributes more when it is far from the average X and has a large residual. Like “outliers”, those observations can move the slope more.

Step 3: construct the standard error

Start with the population expression:

$$SD(\hat{\beta}_1) \approx \frac{1}{\sqrt{n}} \frac{\sqrt{\text{Var}(v_i)}}{\text{Var}(X)}.$$

Plug in the sample variances from the previous two slides:

$$\text{plug-in} \implies SE(\hat{\beta}_1) = \frac{1}{\sqrt{n}} \frac{\sqrt{s_v^2}}{s_X^2}.$$

What the number means

The result estimates the typical sampling error of the slope, in test-score points per additional student for every teacher.

As we'll see later, this is a **heteroskedasticity-robust standard error**.

The small adjustment used by Stata

The sample-variance construction gives

$$\text{plug-in SE} = \sqrt{\frac{n-1}{n}} \frac{\sqrt{\sum_i (X_i - \bar{X})^2 \hat{u}_i^2}}{\sum_i (X_i - \bar{X})^2}.$$

Stata uses the following degrees-of-freedom adjustment:

$$\text{Stata SE}(\hat{\beta}_1) = \sqrt{\frac{n}{n-2}} \frac{\sqrt{\sum_i (X_i - \bar{X})^2 \hat{u}_i^2}}{\sum_i (X_i - \bar{X})^2}.$$

Both estimate the same large-sample uncertainty. Stata's factor accounts for estimating two coefficients. We use its version in the examples below.

For the 420 districts: plug-in SE = 0.517634; Stata robust SE = 0.519489.

Formula convention: Stata, regress, robust variance estimation.

The 3,000-student experiment returns

1. Treat the 420 districts as a fixed teaching population. Its population slope is -2.2798 .
2. Each student draws n districts **uniformly with replacement**. Each draw has probability $1/420$ for every district.
3. Each student reports a slope and its robust standard error.
4. Repeat for $n = 10, 20$, and 50 , with 3,000 students at each size.

What we compare

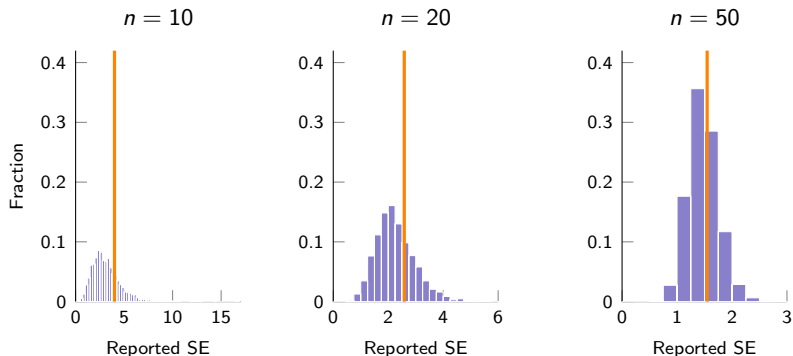
A histogram of the students' *reported SEs* with the fixed *sampling SD of their slope estimator* at that n .

$$\text{Fixed SD: } \text{SD}(\hat{\beta}_1) = \sqrt{E[(\hat{\beta}_1 - E[\hat{\beta}_1])^2]},$$

$$\text{Reported SE: } \text{SE}(\hat{\beta}_1) = \sqrt{\frac{n}{n-2} \frac{\sum_i (X_i - \bar{X})^2 \hat{u}_i^2}{\sum_i (X_i - \bar{X})^2}}.$$

The expectation averages over possible samples of size n . We approximate the fixed SD using 200,000 separate draws at each n .

Reported SEs and the fixed sampling-SD benchmark



Bars: 3,000 reported robust SEs per panel. **Line:** fixed SD benchmark. Horizontal scales differ.

- ▶ The LLN makes the estimated population values more accurate, so reported SEs track the sampling SD more closely as n grows.
- ▶ Larger samples yield smaller SEs and less dispersion in their reported values.
- ▶ Small-sample SEs can substantially underestimate SD of $\hat{\beta}_1$.

What the experiment teaches us about standard errors

n	Fixed SD benchmark	Mean reported SE	SD of reported SE
10	4.008	3.184	1.535
20	2.584	2.270	0.712
50	1.548	1.476	0.274

- ▶ A standard error is itself an estimate, so students disagree about it.
- ▶ At small n , its average can lie below the actual slope dispersion.
- ▶ Larger samples improve the approximation via LLN, but do not make every student's SE equal the benchmark.

Stata: one student's report

```
capture program drop h6_student
program define h6_student, rclass
    syntax, N(integer)
    preserve
    bsample 'n'
    quietly regress testscr str, robust
    return scalar b = _b[str]
    return scalar se = _se[str]
    restore
end
```

- ▶ `bsample` draws with replacement from the loaded population.
- ▶ `_b[str]` is the student's slope.
- ▶ `_se[str]` is the student's reported standard error.

Stata: a class of 3,000 students

Run from the Handout 6 folder, after defining `h6_student`.

```
use "caschool.dta", clear
simulate b=r(b) se=r(se), reps(3000) ///
    seed(38161050): h6_student, n(50)
summarize b se
histogram se, fraction
```

- ▶ The SD of `b` estimates slope dispersion across samples.
- ▶ The mean of `se` summarizes what students report.
- ▶ The SD of `se` measures how much their uncertainty estimates differ.

Reload the population before repeating at $n = 10$ and 20 . The companion do-file runs all three cases and adds the benchmark lines.

Section 5.2

Confidence intervals for the slope

An estimate with room for sampling uncertainty

Suppose two researchers both report $\hat{\beta}_1 = -2.28$:

	Researcher A	Researcher B
Reported SE	0.52	1.50
Estimate $\pm 1.96 \times \text{SE}$	$[-3.30, -1.26]$	$[-5.22, 0.66]$

The same point estimate supports different conclusions when its sampling uncertainty differs.

A confidence interval

A confidence interval surrounds the estimate with a margin that reflects sampling uncertainty.

This margin is determined by

- ▶ Standard error: A larger SE gives a wider interval.
- ▶ Critical value: A higher confidence level gives a larger critical value, which in turn gives a wider interval.

Confidence intervals at 90%, 95%, and 99%

In a large sample, $(\hat{\beta}_1 - \beta_1) / \text{SE}(\hat{\beta}_1)$ is approximately standard normal.
Therefore:

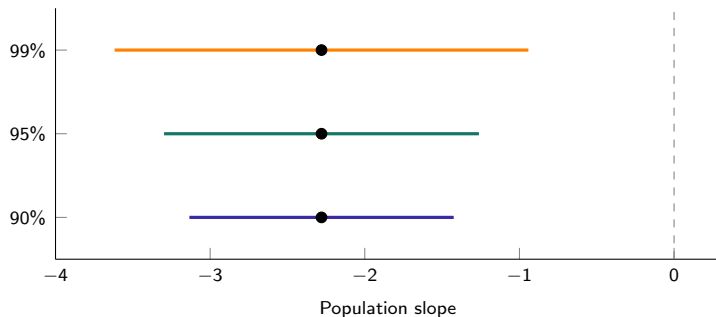
Confidence	Construction	California data
90%	$\hat{\beta}_1 \pm 1.645 \text{ SE}$	$[-3.134, -1.425]$
95%	$\hat{\beta}_1 \pm 1.960 \text{ SE}$	$[-3.298, -1.262]$
99%	$\hat{\beta}_1 \pm 2.576 \text{ SE}$	$[-3.618, -0.942]$

The trade-off

Higher confidence requires a wider interval for the same estimate and SE.

Calculations use the unrounded slope -2.279808 and robust SE 0.519489 . The intervals here use the normal approximation.

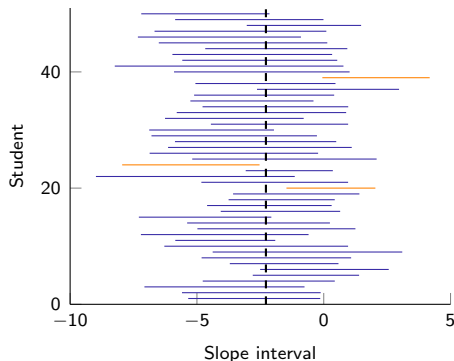
The three confidence intervals in a picture



All three intervals have the same center. Their margins grow with confidence.

Here, even the 99% interval excludes a zero slope.

What 95% confidence means



Each student constructs $\hat{\beta}_1 \pm 1.96 \text{ SE}$.

The population slope stays fixed. The intervals change with the samples.

Orange intervals miss the target.

Shown: the first 50 students
CIs with each of them having a sample size $n = 50$.

Across all 3,000 reports at $n = 50$, coverage is 91.7%. The 95% claim is an approximation, and $n = 50$ does not necessarily guarantee it.

Interpreting a realized confidence interval

For the California regression, the large-sample 95% interval is

$[-3.298, -1.262]$ test-score points per extra student per teacher.

- ▶ Our procedure would contain the fixed population slope in about 95% of repeated large samples.
- ▶ After observing the data, this particular interval either contains the fixed slope or it does not.
- ▶ This is an interval for the *slope*, not a range containing 95% of district test scores.

(Optional) Stata: manual SE, steps 1 and 2

Reload all 420 districts after the sampling exercise.

```
use "caschool.dta", clear
regress testscr str, robust
scalar n = e(N)
scalar b = _b[str]
scalar se_stata = _se[str]
predict uhat if e(sample), residuals
quietly summarize str if e(sample)
generate dx = str-r(mean) if e(sample)
generate dx2 = dx^2
generate vhat2 = (dx*uhat)^2
quietly summarize dx2
scalar Sxx = r(sum)
scalar sX2 = Sxx/(n-1)
quietly summarize vhat2
scalar sv2 = r(sum)/(n-1)
```

(Optional) Stata: manual SE, step 3

Continue immediately after the previous slide:

```
scalar se_plugin = sqrt(sv2/n)/sX2
scalar se_manual = sqrt((n/(n-2))*r(sum)/Sxx^2)
display se_plugin
display se_manual
display se_stata
```

Calculation	Standard error
Sample-variance plug-in	0.517634
Manual calculation with Stata's adjustment	0.519489
regress testscr str, robust	0.519489

S_{xx} is just the saved sum $\sum_i (X_i - \bar{X})^2$. The manual adjusted SE matches Stata to the displayed precision.

(Optional) Stata: confidence intervals

```
foreach level in 90 95 99 {  
    scalar z = invnormal(1-(1-`level'/100)/2)  
    display b-z*se_manual " " b+z*se_manual  
    regress testscr str, robust level(`level')  
}
```

Stata uses a t_{n-2} reference instead of the standard normal for its reported interval. The two become close as n grows. To reproduce its endpoints exactly:

```
scalar c = invttail(e(df_r), .025)  
display b-c*se_stata " " b+c*se_stata
```

95% interval	Critical value	Endpoints
Normal approximation	1.9600	$[-3.298, -1.262]$
Stata, $n = 420$	1.9657	$[-3.301, -1.259]$

This software convention does not make robust inference exact in small samples.

Section 5.1

Testing hypotheses about a regression coefficient

The superintendent's hypothesis

The superintendent wants to know whether class size matters for achievement, in either direction.

$$\underbrace{H_0 : \beta_1 = 0}_{\text{null hypothesis}} \quad \text{versus} \quad \underbrace{H_1 : \beta_1 \neq 0}_{\text{two-sided alternative}} .$$

The decision is to **reject** or **not reject** H_0 .

The error we want to control

A **Type-I error** is rejecting the null when it is true.

- ▶ The **significance level** is the target maximum probability of a Type-I error, approximately in large samples.
- ▶ Researchers choose this level before looking at the test result. Common choices are 10%, 5%, and 1%.

Decision 1: does the 95% interval contain zero?

Decision rule at the 5% significance level

Reject $H_0 : \beta_1 = 0$ if the 95% CI excludes zero. Otherwise, do not reject.

1. Suppose the null is true, so the population slope is zero.
2. A valid 95% interval misses that zero slope in about 5% of repeated samples.
3. Rejecting only when the interval misses zero therefore makes a Type-I error about 5% of the time.

California example

The 95% interval $[-3.30, -1.26]$ excludes zero. We reject the zero-slope hypothesis at the 5% level.

Before sampling, if H_0 is true, this rule falsely rejects in about 5% of repeated large samples. After seeing the data, 5% is not the probability that our particular rejection is wrong.

Decision 2: how many standard errors from zero?

The signed *t*-statistic is

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}.$$

For a two-sided test, distance matters in either direction, so use $|t|$.

$$\begin{aligned} 0 \text{ is outside } \hat{\beta}_1 \pm 1.96 \text{ SE} &\iff |\hat{\beta}_1| > 1.96 \text{ SE} \\ &\iff |t| > 1.96. \end{aligned}$$

$$t = \frac{-2.2798}{0.5195} = -4.389, \quad |t| > 1.96.$$

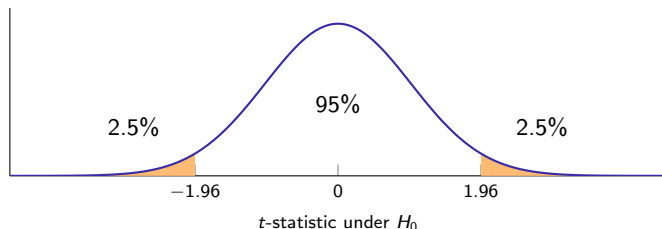
The same decision, read from the *t* column

Reject at 5% if $|t| > 1.96$; otherwise, do not reject. This is equivalent to checking whether the normal-approximation 95% CI excludes zero.

The 5% rejection region under the null

Under H_0 , the t -statistic standardizes the slope just as the z -score standardized the mean in Handout 4.

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} \underset{\sim}{\text{approx.}} N(0, 1).$$



When H_0 holds, about 5% of large-sample t -statistics land in these two tails. Those are the samples in which we reject.

Decision 3: the p -value

The question a p -value answers

If the null hypothesis were true, how likely would we be to obtain a statistic at least as far from zero as the one we observed?

For a two-sided test using the normal approximation:

$$p = \Pr(|Z| \geq |t_{\text{observed}}|), \quad Z \sim N(0, 1).$$

- ▶ A small p -value means the observed distance is unusual under H_0 .
- ▶ A p -value is not the probability that H_0 is true.

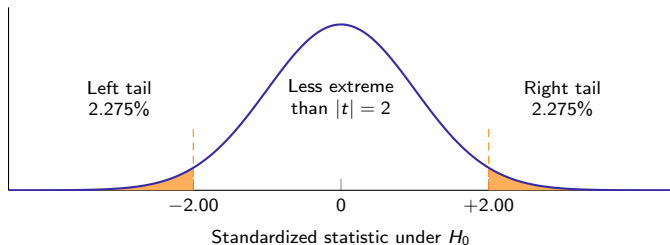
The same decision, read as a probability

Reject at 5% if $p < 0.05$; otherwise, do not reject. With the same reference distribution, this is equivalent to the $|t| > 1.96$ rule.

For the schools regression, the normal-approximation p -value is about 0.0000114. Stata's reference gives about 0.0000145.

The p -value is the area in both tails

For illustration, suppose the observed t -statistic is -2.00 . Under H_0 , use the standard normal approximation.



$$p = \Pr(Z \leq -2) + \Pr(Z \geq 2) \approx 0.0455 = 4.55\%.$$

Both tails count because our alternative is *two-sided*. Since $4.55\% < 5\%$, this example rejects H_0 at the 5% level.

Farther from zero means a smaller p -value

For the actual schools regression, $|t| = 4.389$. The tails beyond -4.389 and $+4.389$ are much smaller: $p \approx 0.0000114$.

Reading the regression output as one argument

```
regress testscr str, robust
```

$n = 420$

	Coefficient	Robust SE	t	$P > t $	95% CI
str	-2.2798	0.5195	-4.39	0.000	[-3.301, -1.259]

1. **CI:** zero lies outside the interval.
2. t : $|t| = 4.389$ exceeds Stata's 5% cutoff, about 1.96.
3. p : the unrounded p -value is below 0.05.

One conclusion

All three reject $H_0 : \beta_1 = 0$ at 5%. They are equivalent when we use the same SE, significance level, and reference distribution.

Stata rounds the displayed p -value to 0.000. The probability is small, not literally zero.

Source: Stock and Watson, `caschoo1.dta`, 420 California school districts.

Stata: the t -statistic and p -value

```
use "caschool.dta", clear
regress testscr str, robust
scalar t = _b[str]/_se[str]
display t
display 2*normal(-abs(t))
display 2*ttail(e(df_r),abs(t))
```

The first probability uses the large-sample normal approximation. The second reproduces the reference used in Stata's coefficient table. *The slope is statistically different from zero at the 5% level.*

Statistical significance, magnitude, and causality

- ▶ **Statistical significance:** the estimated slope is far from zero relative to its sampling uncertainty.
- ▶ **Magnitude:** one more student per teacher is associated with about 2.28 *fewer* test-score points. Whether that change matters depends on the score scale, the policy change, and its costs.
- ▶ **Causality:** district income may affect both STR and achievement. Such omitted factors could explain the observed association even if changing class size itself has no effect.

What the test establishes

Evidence against a zero population slope does not by itself establish a large policy benefit or a causal effect.

Section 5.3

Regression when the regressor is binary

A simpler comparison: two groups of districts

Define the **Class Size Indicator**:

$$CSI_i = \begin{cases} 0 & STR_i < 20, \\ 1 & STR_i \geq 20. \end{cases}$$

Group	Districts	Average test score
Smaller STR, $CSI = 0$	238	657.35
Larger STR, $CSI = 1$	182	649.98

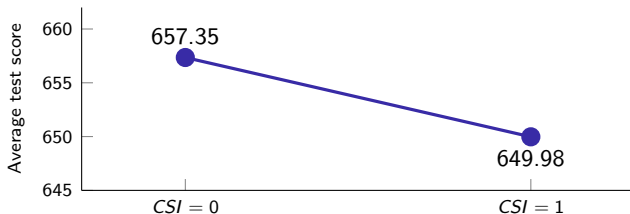
The difference of Test Scores, *larger STR minus smaller STR*, is

$$649.98 - 657.35 = -7.37 \text{ points.}$$

We use $STR \geq 20$ for the group coded one, including the equality.

Source: Stock and Watson, `caschoo1.dta`, 420 California school districts.

The regression connects the two group averages



$$\widehat{\text{TestScore}} = 657.35 - 7.37 \text{ CSI}.$$

- ▶ The intercept, 657.35, is the mean test score for $\text{CSI} = 0$.
 - ▶ The slope, -7.37 , is the mean for $\text{CSI} = 1$ minus the mean for $\text{CSI} = 0$.
- The next two slides explain these relationships using conditional means.

Why the intercept is the mean for the zero group

Write

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{CSl}_i + u_i.$$

For districts with $\text{CSl}_i = 0$:

$$\text{TestScore}_i = \beta_0 + u_i.$$

Take the average within that group:

$$E[\text{TestScore}_i \mid \text{CSl}_i = 0] = \beta_0 + E[u_i \mid \text{CSl}_i = 0] = \beta_0.$$

Using Least Square Assumption 1

The mean regression error is zero within each group:

$$E[u_i \mid \text{CSl}_i = 0] = E[u_i \mid \text{CSl}_i = 1] = 0.$$

Why the slope is a difference in means

For districts with $CSl_i = 1$, the same reasoning gives

$$E[\text{TestScore}_i \mid CSl_i = 1] = \beta_0 + \beta_1 + E[u_i \mid CSl_i = 1] = \beta_0 + \beta_1.$$

Subtract the mean for the zero group:

$$\beta_1 = E[\text{TestScore}_i \mid CSl_i = 1] - E[\text{TestScore}_i \mid CSl_i = 0].$$

In a summary, for any binary regressor X_i :

$$\begin{aligned}\beta_0 &= E[Y_i \mid X_i = 0], \\ \beta_1 &= E[Y_i \mid X_i = 1] - E[Y_i \mid X_i = 0].\end{aligned}$$

A binary-regressor model with an intercept can represent both group means exactly. This descriptive identity does not make the group difference causal.

Stata: group means and the binary regression

```
use "caschool.dta", clear
generate CSI = (str>=20) if !missing(str)
tabstat testscr, by(CSI) statistics(n mean sd)
regress testscr CSI, robust
graph bar (mean) testscr, over(CSI) ///
    ytitle("Average test score")
```

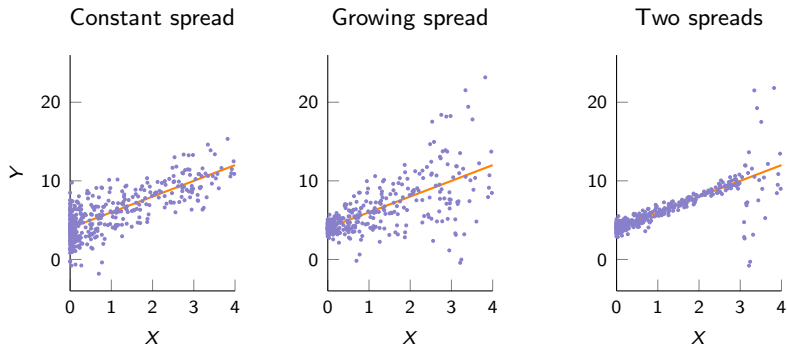
	Estimate	Robust SE
Intercept	657.35	1.255
<i>CSI</i>	-7.37	1.824

The slope's 95% Stata CI is $[-10.957, -3.788]$. The two groups differ statistically. Interpreting the gap causally requires A1 for a credible causal model; the group-means identity alone is not enough.

Section 5.4

Heteroskedasticity and homoskedasticity

Same average relationship, different amounts of scatter



Same 400 X values and underlying noise in all three simulations.

- ▶ All three populations center on the orange line, $E[Y | X] = 4 + 2X$. A1 holds by construction.
- ▶ The left has equal spread at each x (homoskedasticity). The middle and right have changing spread (heteroskedasticity).

(Optional) Stata: constructing the three examples

```
clear
set obs 400
set seed 38163000
generate x = 4*runiform()^3
generate noise = rnormal()
generate y_equal = 4+2*x+2*noise
generate y_fan = 4+2*x+(.5+1.5*x)*noise
generate y_groups = 4+2*x+(.5+5*(x>=3))*noise
twoway (scatter y_fan x) ///
      (function y=4+2*x, range(0 4)), legend(off)
```

noise is independent standard normal noise. Most X values are small, while the largest errors occur at the few large X values. This makes the choice of SE matter more.

Replace `y_fan` with the other outcomes `y_equal` or `y_groups` to draw the other panels.

The definitions describe conditional variance

In the model $Y_i = \beta_0 + \beta_1 X_i + u_i$:

Homoskedasticity	Heteroskedasticity
$\text{Var}(u_i \mid X_i = x)$ is constant in x .	$\text{Var}(u_i \mid X_i = x)$ changes with x .
Equal error spread at every x .	Different error spread at different x .

In our simulations the conditional variances are, respectively,

$$4, \quad (.5 + 1.5x)^2, \quad \begin{cases} 0.25 & x < 3, \\ 30.25 & x \geq 3. \end{cases}$$

A1 can hold in every case

$E[u_i \mid X_i] = 0$ restricts the mean. It does not require a constant variance.

Why the variance separates under homoskedasticity

Recall $v_i = (X_i - E[X])u_i$. Imagine sorting observations into groups that share the same value $X = x$.

1. **Within a group**, $x - E[X]$ is fixed. Multiplying every error by that number multiplies its variance by $(x - E[X])^2$.
2. **Across groups**, homoskedasticity gives the same error variance $\text{Var}(u)$ in every group. A1 gives a zero mean for v_i in every group, so group means add no extra variation.
3. **Average over groups**, weighting by their population shares. The common $\text{Var}(u)$ factors out, and the average squared distance $(X - E[X])^2$ is $\text{Var}(X)$.

$$\text{Var}(v_i) = \text{Var}(X) \text{Var}(u).$$

What homoskedasticity simplifies

The general slope-variance formula contains the variance of v_i . Under homoskedasticity we have just shown

$$\text{Var}(v_i) = \text{Var}(X) \text{Var}(u).$$

Substitute into the general formula:

$$\text{Var}(\hat{\beta}_1) \approx \frac{1}{n} \frac{\text{Var}(u)}{\text{Var}(X)}, \quad \text{SD}(\hat{\beta}_1) \approx \frac{1}{\sqrt{n}} \sqrt{\frac{\text{Var}(u)}{\text{Var}(X)}}.$$

More error variation reduces precision. More variation in X improves precision, holding the error variance fixed.

Two standard-error formulas

For one regressor and an intercept, Stata's homoskedasticity-only formula is

$$SE_{\text{homo}}(\hat{\beta}_1) = \sqrt{\frac{\sum_i \hat{u}_i^2 / (n-2)}{\sum_i (X_i - \bar{X})^2}} = \frac{SER}{\sqrt{(n-1)s_X^2}}.$$

It estimates $\frac{1}{\sqrt{n}} \sqrt{\text{Var}(u) / \text{Var}(X)}$ and requires homoskedasticity.

The robust formula keeps the squared errors attached to their X values:

$$SE_{\text{robust}}(\hat{\beta}_1) = \sqrt{\frac{n}{n-2} \frac{\sum_i (X_i - \bar{X})^2 \hat{u}_i^2}{[\sum_i (X_i - \bar{X})^2]^2}}.$$

Both formulas use **the same OLS slope and residuals**. They differ in how they estimate its sampling uncertainty.

(Optional) Stata: manually comparing the two SEs

Use the school data and the saved n , Sxx , $uhat$, and $vhat2$ from the manual-SE slides in Section 5.2.

```
generate uhat2 = uhat^2
quietly summarize uhat2
scalar se_homo = sqrt((r(sum)/(n-2))/Sxx)
quietly summarize vhat2
scalar se_robust = sqrt((n/(n-2))*r(sum)/Sxx^2)
regress testscr str
display se_homo " " _se[str]
regress testscr str, robust
display se_robust " " _se[str]
```

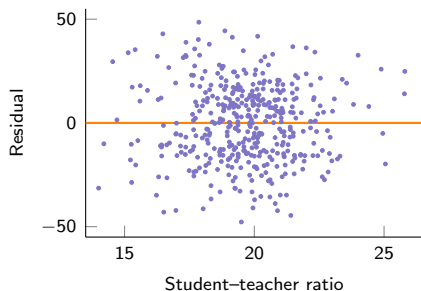
	Manual SE	Stata SE
Homoskedasticity-only	0.479826	0.479826
Heteroskedasticity-robust	0.519489	0.519489

The estimates agree; the uncertainty can differ

Example	OLS slope	Homoskedasticity-	
		only SE	Robust SE
California schools	-2.280	0.480	0.519
Constant spread	1.983	0.086	0.084
Growing spread	1.913	0.107	0.176
Two spreads	1.861	0.070	0.142

- ▶ The robust SE is about 1.65 times as large for growing spread and 2.04 times as large for two spreads.
- ▶ The high-variance observations are far from the mean of X , where they strongly influence the slope. The homoskedasticity-only formula treats their error variance as common to all observations.
- ▶ The fitted line stays the same. The CI, t -statistic, and p -value can change.
- ▶ Robust SEs can be larger or smaller. Robust does not mean always larger.

Stata: looking for changing residual spread



```
use "caschool.dta", clear
regress testscr str, robust
predict uhat, residuals
scatter uhat str, yline(0)
```

Does the vertical spread look constant? Sparse observations at extreme STR make visual judgments difficult.

A residual plot can suggest patterns. It cannot prove homoskedasticity or the least square assumptions.

Source: Stock and Watson, `caschool1.dta`, 420 California school districts.

Our default: heteroskedasticity-robust standard errors

The course rule

Use robust standard errors for our regressions with independent observations, whether or not the scatterplot looks homoskedastic.

```
regress testscr str, robust  
regress testscr str, r
```

- ▶ Both commands request the same robust SE.
- ▶ Robust SEs are valid under **both** homoskedasticity and heteroskedasticity, given our other assumptions.
- ▶ They do not fix omitted factors correlated with X , dependence between observations, or unreliable small-sample approximations.

When the large-sample approximation is useful

The confidence intervals and tests in this handout rely on a **large-sample approximation**.

- ▶ The textbook's rough guide is a sample size above about 50.
- ▶ This is a starting point, not a guarantee. Outliers, limited variation in X , or very uneven binary groups can require more observations.
- ▶ Our $n = 10, 20, 50$ experiment shows why the quality of the approximation matters.

Before interpreting a result

Identify the population and slope, check the sampling and regression assumptions, and use the robust SE to express the remaining sampling uncertainty.

We leave the additional OLS efficiency theory and small-sample theory in Sections 5.5–5.6 outside this handout.

New notions and methods in Handout 6

New notions

- ▶ Standard error: estimated sampling SD of a coefficient.
- ▶ Confidence interval: a procedure with repeated-sample coverage.
- ▶ Null and alternative hypotheses.
- ▶ Type-I error, significance level, t -statistic, and p -value.
- ▶ Binary regressor and difference in group means.
- ▶ Constant versus changing conditional error variance.

New methods

- ▶ Replace u_i by $\hat{u}_i$ and population quantities by sample quantities.
- ▶ Form a 95% CI: $\hat{\beta}_1 \pm 1.96 \text{ SE}$.
- ▶ For $H_0 : \beta_1 = 0$, reject at 5% when the CI excludes zero, $|t| > 1.96$, or $p < 0.05$.
- ▶ For binary X_i , the slope is the mean for $X_i = 1$ minus the mean for $X_i = 0$.
- ▶ In Stata, use `regress Y X, robust`.

The normal-approximation rules require a sufficiently large sample and the stated assumptions. Statistical significance alone does not establish causality.